

The Cost-Effectiveness of Alternative Strategies for the Diagnosis and Treatment of Moderate Angina

Patient Cohort: Moderate Typical Angina Pain

Perspective: Sheffield Cardiology Department

Abstract

Background: Angina pectoris, a symptom of coronary artery disease (CAD), requires clinical evaluation for diagnosis and treatment. For male patients aged 55 years old with moderate typical angina, clinicians must decide whether to perform an exercise stress test (EST), an invasive procedure called coronary angiography (ANG), or no diagnostic test (No Test), and then select either no treatment (NT), medical management (MM), or coronary artery bypass graft surgery (CABG).

Methods: A decision-analytic tree model was used to evaluate 15 different diagnostic and treatment strategies from the perspective of a Sheffield cardiology department. The model has 46 terminal nodes, and for those EST and ANG-based approaches, has a specific treatment for test-positive (T+) and test-negative (T-) patients. The Rosser matrix (Williams, 1985) was used to obtain Quality Adjusted Life Years (QALYs) for each health status, with a 3.5% discount. A comparison of the strategies was made based on net monetary benefit (NMB) and incremental cost-effectiveness ratios (ICER). Our sensitivity analyses examined a range of variables, from the value of perfect care or diagnostic information to the likelihood of procedural death in ANG and CABG cases. We also considered discount rates ranging from 0 to 6 percent and WTP thresholds from £53 to £35,000 per QALY.

Results: At WTP=£35,000/QALY and a 3.5% discount rate for health and costs, of 15 strategies, ten were strictly dominated, two were extendedly dominated, and the remaining for the frontier comparison were: the baseline strategy, (EST,t-,NT);(EST,t+,MM) with an ICER of £908/QALY, and the cost-effective winner (ANG,t-,NT);(ANG,t+,CABG) with an ICER of £2,554/QALY. The WTP threshold sensitivity analysis showed that at the very low threshold, (EST,t-,NT);(EST,t+,MM) emerged as the winner. Results were robust to changes in the discount rate. Also, they were robust to quite a great extent up to 20 times that of the ANG/CABG mortality rate, compared to the baseline of SVD, which means it would have to be unrealistically high procedural mortality to alter the optimal strategy. Perfect care is valued at 114 times that of perfect diagnosis (£28,153 vs £247 NMB) in the value of perfect information and perfect care sensitivity analysis.

1. Introduction

1.1 Background and Motivation

Angina pectoris is a symptom of coronary artery disease (CAD) that creates a significant burden on health systems. Patients with typical angina may have an exercise stress test (EST), which is noninvasive but not always reliable due to false positive and false negative probabilities, or coronary angiography (ANG), which is invasive but gives clear anatomical details as it is considered the gold standard. Patients after diagnosis have three options of treatments: No treatment (NT), Medical management (MM) for life, or coronary artery bypass graft surgery (CABG) followed by lower cost of MM if the surgery is successful. Choosing the best option involves weighing cost in pounds against health outcomes, using a per-QALY (Quality Adjusted Life Years) analysis with Net Monetary Benefit (NMB) and the Incremental Cost-Effectiveness Ratio (ICER), against a willingness-to-pay (WTP) limit set by the National Institute for Health and Care Excellence (NICE).

1.2 Research questions

This report answers the following questions for male patients aged 55 with moderate typical angina referred to a Sheffield cardiology department: What are the expected costs and QALYs for each of the fifteen diagnostic and treatment strategies? -Which strategies are on the cost-effectiveness frontier, and what are their ICERs?

Even though this analysis is mainly concerned with the Sheffield/UK perspective, it also considers that the results would be used in different settings within the UK, with some uncertainty about the procedural performance. It aims to answer: How sensitive are the results to procedural death probabilities for ANG and CABG, and at what mortality levels would the optimal strategy change across different surgical settings?

Moreover, many countries use the UK's prices or NICE health technology assessments as benchmarks in their External Reference Pricing (ERP) systems with different WTPs. Given that, the next question is: How does the recommended strategy change across WTP thresholds, especially below the NICE range? Ruggeri, K., & Nolte, E. (2013).

And ultimately, what is the value of achieving perfect diagnostic information and perfect care?

1.3 Structure of the Report

The data and the methodologies employed are detailed in Section 2, and any necessary assumptions or simplifications are explained in the text as they apply. Section 3 then presents the results from the primary, or 'base case', scenario. Section 4 presents the findings of the sensitivity analyses. Finally, Section 5 discusses the findings, policy implications, and key assumptions.

2. Data and Methods

2.1 Patient Cohort and Perspective

The cohort comprises male patients aged 55 with moderate typical angina, a prior probability of coronary artery disease of 60% [$P(\text{CAD})=0.6$], good left ventricular function, and no prior myocardial infarction. Therefore, coronary pathology is simplified to three mutually exclusive states: no disease (ND), single-vessel disease (SVD), and multi-vessel disease (MVD), and is modeled. As a limitation, a more granular classification, such as left main stem disease or three-vessel disease, would better reflect clinical heterogeneity and could alter the relative benefit of CABG versus MM. Moreover, the results may not be externally valid for women, older patients, those with impaired left ventricular function, or those with prior MI.

Life expectancy varies by disease state and, for MVD patients, by treatment received; CABG extends life expectancy by 5-10 years relative to MM or NT. However, within each treatment-

disease combination, life expectancy is fixed and does not respond to individual patient characteristics or disease progression over time. A Markov model would capture year-by-year transitions more accurately, though the current model already reflects most of the treatment benefit through differential life expectancy for MVD patients and will capture the diagnostic tests (ANG, EST, No Test) benefit perfectly, given that the tests are one-time, unlike the treatments.

2.2 Strategies Compared

Fifteen strategies are evaluated, combining diagnostic approaches (No Test, EST, direct ANG) with treatment options (NT, MM, CABG). EST-based strategies specify distinct treatments for T+ and T- patients. Table 1 lists all fifteen strategies.

Strategy	T+ Treatment	T- Treatment
No Test, NT	No treatment	--
No Test, MM	Medical management	--
(EST,t-,NT);(EST,t+,NT)	NT	NT
(EST,t-,NT);(EST,t+,MM)	MM	NT
(EST,t-,MM);(EST,t+,MM)	MM	MM
(EST,t-,MM);(EST,t+,NT)	NT	MM
(EST,t-,NT);(EST,t+,ANG,t+,NT)	ANG then NT	NT
(EST,t-,NT);(EST,t+,ANG,t+,MM)	ANG then MM	NT
(EST,t-,MM);(EST,t+,ANG,t+,MM)	ANG then MM	MM
(EST,t-,MM);(EST,t+,ANG,t+,NT)	ANG then NT	MM
(EST,t-,NT);(EST,t+,ANG,t+,CABG)	ANG then CABG	NT
(EST,t-,MM);(EST,t+,ANG,t+,CABG)	ANG then CABG	MM
(ANG,t-,NT);(ANG,t+,NT)	NT (ND confirmed)	NT
(ANG,t-,NT);(ANG,t+,MM)	MM (disease confirmed)	NT (ND)
(ANG,t-,NT);(ANG,t+,CABG)	CABG (disease confirmed)	NT (ND)

Table 1: Fifteen strategies evaluated.

As noted in Table 1, treatment is applied uniformly regardless of the disease state revealed by ANG, which is a simplification of the model. A fully disaggregated model would consider disease-state-specific pathways. For example, instead of (ANG,t-,NT);(ANG,t+,CABG) and (ANG,t-,NT);(ANG,t+,MM), it would generate four strategies:

1. (ANG,t-,NT);(ANG,t+,SVD,CABG);(ANG,t+,MVD,CABG)
2. (ANG,t-,NT);(ANG,t+,SVD,MM);(ANG,t+,MVD,MM)
3. (ANG,t-,NT);(ANG,t+,SVD,MM);(ANG,t+,MVD,CABG)
4. (ANG,t-,NT);(ANG,t+,SVD,CABG);(ANG,t+,MVD,MM)

This would approximately double the number of strategies. The current model, therefore, would underestimate the true value of perfect ANG information, since the optimal mixed strategy would extract greater benefit from confirmed disease status.

2.3 Decision Analytic Model

A decision tree with 46 terminal nodes estimates expected costs and QALYs for each strategy. The names for the strategies are formulated by the intervention choices (treatment or diagnostic tool) at each decision node, for example: (EST,t-,NT) and (EST,t+,MM) denote strategies in which a negative EST leads to NT and a positive EST leads to MM. However, for each strategy, E[U] and E[C] were calculated by multiplying each utility and cost at each terminal node by its path probability. The path probability is the product of all branch probabilities along that path. Across all terminal nodes within each strategy, path probabilities sum to 1.0, confirming the model's internal consistency. However, at decision nodes, the Net monetary benefit was the metric to determine which candidate to advance to the next stage (to the left).

EST has a sensitivity of 0.85 and a specificity of 0.90. ANG is a prerequisite for CABG. Although it is assumed to provide perfect information, revealing the true disease state (SVD, MVD, h) with certainty, in practice, some diagnostic uncertainty remains, which can slightly overstate the benefit of ANG-based strategies and limit the model accuracy.

As noted in *Table 1*, the ANG test can follow a positive EST or be used directly. When ANG follows a positive EST, the posterior probability of disease replaces the prior. Applying Bayes' theorem: $P[s|T+] = (0.85 \times 0.6)/0.55 = 0.927$; $P[h|T+] = 0.073$. Because $P(SVD|s) = 0.3$ and $P[MVD|s] = 0.7$, the disease-state probabilities for ANG after a positive EST are: $P(SVD|T+) = (0.927 \times 0.3) = 0.278$; $P(MVD|T+) = (0.927 \times 0.7) = 0.649$; $P[h|T+] = 0.073$. However, for direct ANG: $P(SVD) = 0.18$, $P(MVD) = 0.42$, $P[h] = 0.40$.

ANG is not used after a negative EST when $P(s|T-) = 0.200$, which is one of the model's assumptions/simplifications. However, as this model demonstrates, the incremental health gain from offering MM to T- patients is negligible in the base case, suggesting that ANG after T- would add little additional benefit, consistent with clinical guidelines, which state that negative stress testing rules out the need for anatomic testing, including invasive coronary angiography (Gulati et al., 2021).

Moreover, that failed CABG is assumed to result in outcomes (utility) equivalent to those of MM without surgery. In practice, failed surgery (CABG) can carry higher morbidity than MM alone, which would reduce the expected utility of surgical strategies. A simplified version of the decision tree is presented in Figure 1. The full model is in Appendix D.



Figure 1: Simplified decision tree

2.4 Probabilities

Parameter	Value	Notes
Prior P(s)	0.600	Case study
Prior P(h)	0.400	Case study
P(SVD s)	0.300	Case study
P(MVD s)	0.700	Case study
EST Sensitivity	0.850	Case study
EST Specificity	0.900	Case study
PPV - P(s T+)	0.927	Derived (Bayes)
NPV - P(h T-)	0.800	Derived (Bayes)
P(SVD T+) for ANG after EST	0.278	0.927×0.300
P(MVD T+) for ANG after EST	0.649	0.927×0.700
P(death,ANG,h)	0.0002	Case study
P(death,ANG,SVD)	0.0003	Case study
P(death,ANG,MVD)	0.0010	Case study
P(death,CABG,SVD)	0.030	Case study
P(death,CABG,MVD)	0.080	Case study
P(good response,CABG,SVD)	0.850	Case study
P(good response,CABG,MVD)	0.700	Case study

Table 2: Key model probabilities.

As mentioned in *Table 2*, CABG success probability is assumed to be constant within disease states (0.85 for SVD, 0.70 for MVD), regardless of individual patient characteristics such as comorbidities.

2.5 Health Outcomes: QALYs

Health outcomes were measured in QALYs using the Rosser matrix (Williams, 1985). The Rosser matrix is an older preference-based instrument; current HTA practice favours the EQ-5D-5L, so this instrument is likely to be biased. Each patient had a sequence of health states in distress for each year of remaining life expectancy. The annual utility was discounted at 3.5% per HM Treasury Green Book guidance (HM Treasury, 2022).

Case	Disease State	Treatment	Discounted QALYs	Life Years
SVD.CABG	SVD	CABG (good response)	8.5123	10
SVD.MM	SVD	Medical management or CABG (bad response)	8.3426	10
SVD.NTA	SVD	No treatment	6.1517	10
MVD.CABG	MVD	CABG (good response)	8.3069	10
MVD.MM	MVD	Medical management or CABG (bad response)	4.2258	5
MVD.NTA	MVD	No treatment	3.3237	5
h.NT or h.MM	(H) healthy	No treatment / MM	11.8002	15

Case	Disease State	Treatment	Discounted QALYs	Life Years
Death	--	--	0.000	0

Table 3: Discounted QALY values

2.6 Costs

Costs present the direct costs according to the Sheffield cardiology department.

Annual MM: £157/year without surgery or post-unsuccessful CABG, £107/year post a successful CABG.

CABG: £3,527, ANG: £668, EST: £53, as a one-off cost, undiscounted.

annual MM costs discounted at 3.5%, the NICE UK discounting rate. And *Table 4* summarizes the cost items. The MM cost is assumed to be constant throughout the treatment duration, which is a limitation because, in practice, through disease progression and cost inflation, the treatment cost typically increases due to the increased intensity of the treatment or to price changes in the medication over time, thereby not capturing and underestimating the total MM cost. Another limitation of this model is that it does not consider the cost of productivity loss after the surgery and the costs of follow-up, re-interventions, and emergency admissions of CABG and MM strategies

Cost Item	Value	Frequency	Discounted Total
EST	£53.00	Once	£53.00
ANG	£668.00	Once	£668.00
CABG surgery	£3,527.00	Once	£3,527.00
MM post-CABG SVD or MVD (10 yrs)	£107/yr	Annual	£921.02
MM without surgery SVD (10 yrs) or post-unse-CABG	£157/yr	Annual	£1,351.41
MM without surgery MVD (5 yrs) or post-unse-CABG	£157/yr	Annual	£733.67
MM without surgery H (15 yrs)	£157/yr	Annual	£1,871.52

Table 4: Discounted costs (3.5% discount rate).

Unse: unsuccessful; H: healthy

2.7 Decision Rule

Strategies were compared using $ICER = \Delta C / \Delta U$ and $NMB = (WTP \times E[U]) - E[C]$. Strictly dominated and extendedly dominated strategies were excluded. Both costs and effects are discounted at 3.5%, and the willingness-to-pay (WTP) = £35000/QALY, based on the most recent NICE threshold, was used in the base case.

A range of deterministic sensitivity analyses was conducted to assess the validity of the base-case analysis's conclusion and to explore different scenarios, primarily using the NMB metric for logistical convenience over ICER, due to the need to manually exclude strictly and extendedly dominated strategies. The WTP sensitivity analysis was tested over a range from £53/QALY to £35000/QALY, while the discount rate remained at the base case, to identify crossover points among the 15 strategies. A line graph was used to present the crossovers. Second, the discount rate for both health and cost was tested over a range of 0 to 6% together, while the WTP threshold was held at the base case, and $E(c)$ and $E(u)$ were used with NMB to demonstrate the effect of the discount rate on the winning strategy. Moreover, the likelihood of death with both (ANG) and (CABG) was tested for each illness and, collectively, across all illnesses, to account for differences in surgical skill across settings, to assess the acceptable safety margin for each procedure, and to identify crossovers between the strategies. Finally, the value of achieving perfect diagnostic information and perfect care was calculated to quantify the

maximum potential gain from improving EST accuracy and surgical performance, and to assess whether the winning strategy would change.

3. Base Case Results

3.1 Expected Costs and QALYs

Table 5 presents E[U], E[C], and NMB at WTP=35,000 for all fifteen strategies. Ten strategies were strictly dominated, and two were extendedly dominated, leaving three on the frontier in Table 6.

Strategy	E(U)	E(C)	NMB (35k)
No Test, NT	7.2233	0.00	252,816
(EST,t-,NT);(EST,t+,NT)	7.2233	53.00	252,763
(EST,t-,NT);(EST,t+,ANG,t+,NT)	7.2217	420.40	252,341
(EST,t-,NT);(EST,t+,MM)	7.8806	596.55	275,224
(ANG,t-,NT);(ANG,t+,NT)	7.2206	668.00	252,054
(EST,t-,MM);(EST,t+,NT)	7.3393	809.46	256,066
(EST,t-,NT);(EST,t+,ANG,t+,MM)	7.8786	888.76	274,862
(EST,t-,MM);(EST,t+,ANG,t+,NT)	7.3377	1,176.86	255,644
(ANG,t-,NT);(ANG,t+,MM)	7.9934	1,219.01	278,550
No Test, MM	7.9966	1,300.00	278,580
(EST,t-,MM);(EST,t+,MM)	7.9966	1,353.00	278,527
(EST,t-,MM);(EST,t+,ANG,t+,MM)	7.9946	1,645.22	278,165
(EST,t-,NT);(EST,t+,ANG,t+,CABG)	8.6784	2,647.73	301,097
(ANG,t-,NT);(ANG,t+,CABG)	8.9344	3,288.39	309,415
(EST,t-,MM);(EST,t+,ANG,t+,CABG)	8.7944	3,404.19	304,401

Table 5: Cost-effectiveness results at WTP=35,000/QALY.

3.2 Cost-Effectiveness Frontier and ICERs

Strategy	E(U)	E(C)	Inc. QALYs	Inc. Cost	ICER (per QALY)	NMB
No Test, NT	7.2233	0.00	--	--	Baseline	252,816
(EST,t-,NT);(EST,t+,MM)	7.8806	596.55	0.6573	596.55	908	275,224
(ANG,t-,NT);(ANG,t+,CABG)	8.9344	3,288.39	1.0538	2,691.84	2,554	309,415

Table 6: Cost-effectiveness frontier.

The ICER for (EST,t-,NT);(EST,t+,MM) compared to (No Test, NT) is £908/QALY. This shows the high diagnostic yield of EST, where $P(s)=0.6$. The strategy (ANG,t-,NT);(ANG,t+,CABG) is the most cost-effective strategy under the NICE threshold and a 3.5% discount rate. With ICER compared to EST, MM is £2,554/QALY, which is well below the NICE threshold range. Importantly, this frontier table is identical to a simpler model that assumed NT only after EST, T- but with different ICERS. This confirms that the option to follow T- with MM does not change the

main conclusions in the base case. This may be because $P(s|T^-)$ equals 0.200. The QALY gain from offering MM to this 20% is very small compared to the extra cost.

4. Sensitivity Analysis

4.1 Willingness-to-Pay Threshold

The Net Monetary Benefit (NMB) was determined for all fifteen strategies, using a WTP threshold from £53 to £35,000 per QALY. The £53/QALY lower limit is the cost of (EST). NMBs across this entire range are displayed in Figure 2, and the more detailed section of the graph makes the two points at which the best strategy shifts very apparent in Figure 3 for only the relevant strategies (that had a chance of winning at some point). Moreover, as shown in Table 7, most of the changes occur between £0 and £2554; beyond that, the optimal strategy remains the same as in the base case, indicating dominance at $WTP=£2554$ and above.

WTP Threshold	Optimal Strategy	Interpretation
Below £908	No Test, NT	No intervention is cost-effective; QALY gains do not justify any expenditure at this threshold
908 to £2,554	(EST,t-,NT);(EST,t+,MM)	EST with MM for T+ offers the best value
Above £2,554	(ANG,t-,NT);(ANG,t+,CABG)	Full invasive pathway is cost-effective; winner at £2554 and beyond

Table 7: Optimal strategy by WTP threshold.

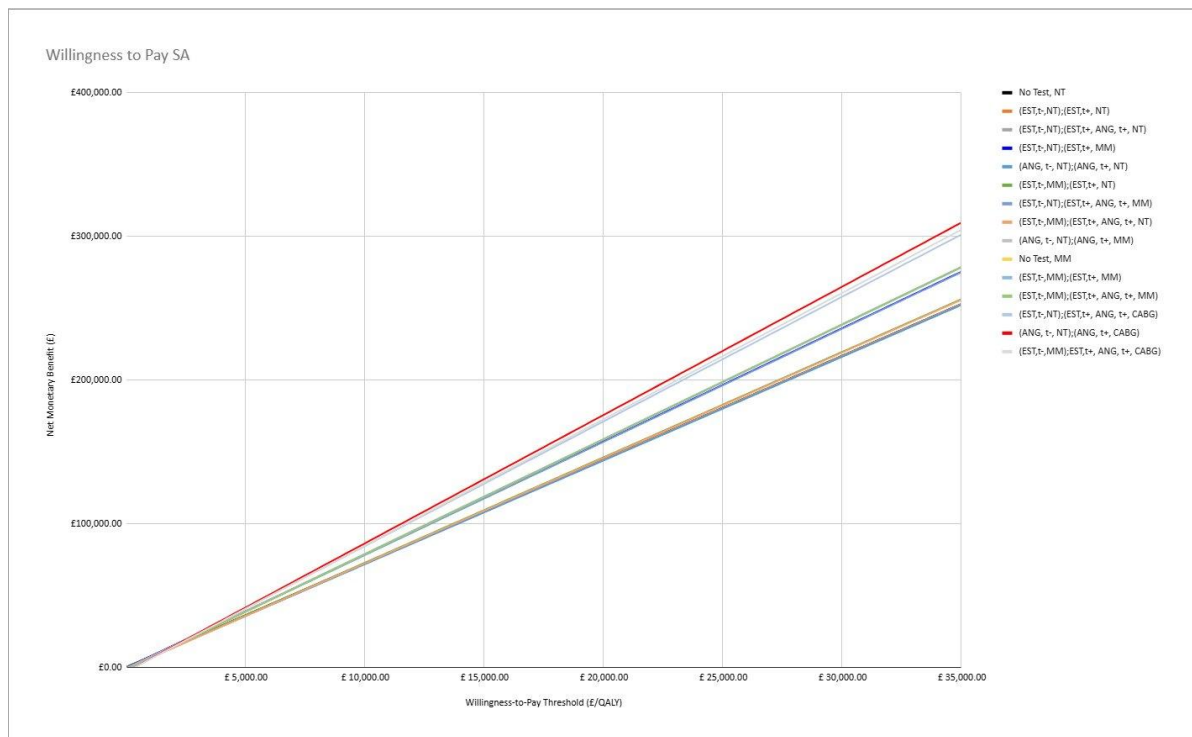


Figure 2: NMB for all fifteen strategies across WTP thresholds £53 to £35,000/QALY.

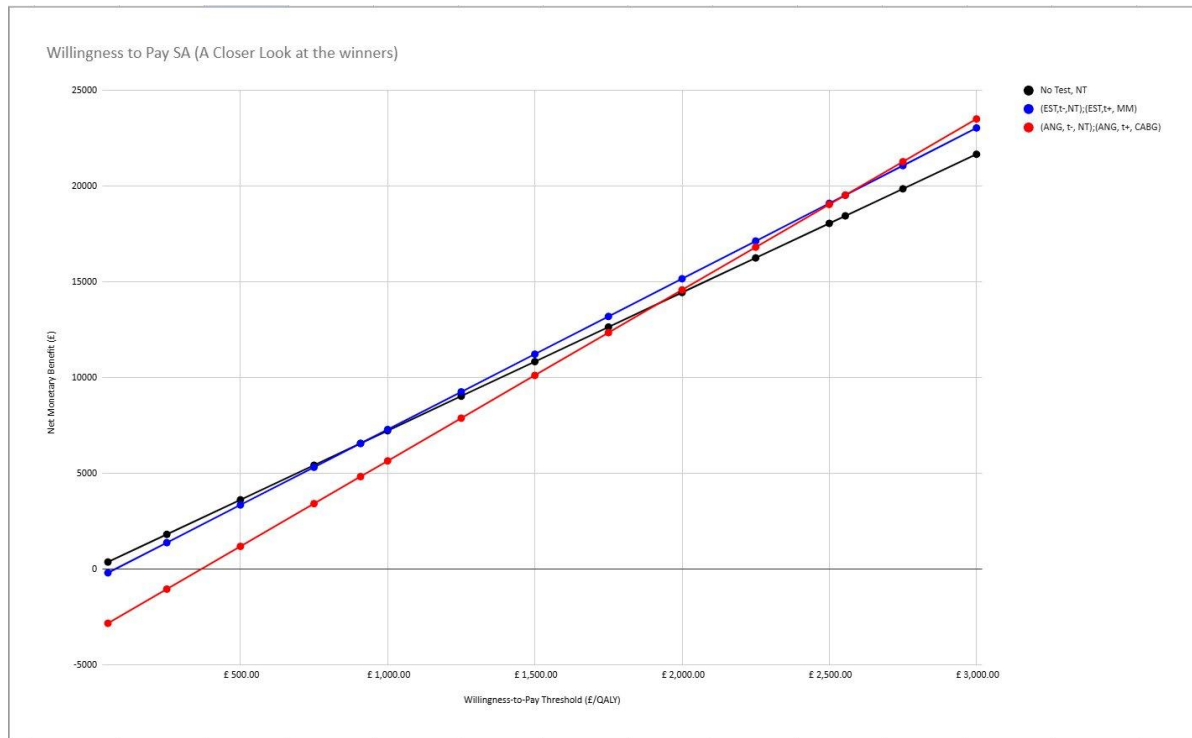


Figure 3: Close-up NMB chart: three frontier strategies with crossover points.

Results are not very sensitive to the choice of threshold because the winning strategy is well below the base-case WTP threshold, given the low cost. Most of the relevant effects of this sensitivity analysis occur at the lower bound of the range, specifically between £0 and £2554. However, as mentioned in Figure 3 and Table 7, above £2,554/QALY, the optimal strategy, (ANG,t-,NT);(ANG,t+,CABG), remains cost-effective regardless of how high the threshold rises, because it has the highest E(U) among all 15 strategies, indicating stability across most of the range.

4.2 Discount Rate

Another sensitivity analysis is conducted by varying the discount rate from 0% to 6%, applied simultaneously to costs and effects, which did not change the optimal strategy at WTP=35,000.

As shown in Table 8, the NMB is reduced by an increasing discount rate, while the ICER increased until £3024.12 at 6%. Figure 4 importantly shows that the optimal strategy E(U) is much more sensitive to this analysis than E(C). This is due to the nature of the strategy: the largest share of the cost comes from the surgery itself (a one-time procedure in year zero, not subject to discounting), whereas a smaller share comes from the MM after surgery. Therefore, most of the discounting effect impacts E(U), the benefit, rather than the cost.

Discount Rate	Winner	NMB	E(U)	E(C)
0%	(ANG,t-,NT);(ANG,t+,CABG)	373,477	10.767	3,362
1%	(ANG,t-,NT);(ANG,t+,CABG)	353,031	10.182	3,340
2%	(ANG,t-,NT);(ANG,t+,CABG)	334,396	9.649	3,318
3.5%	(ANG,t-,NT);(ANG,t+,CABG)	309,415	8.934	3,288
5%	(ANG,t-,NT);(ANG,t+,CABG)	287,510	8.308	3,262
6%	(ANG,t-,NT);(ANG,t+,CABG)	274,383	7.932	3,245

Table 8: Discount rate sensitivity (WTP=35,000). Base case highlighted.

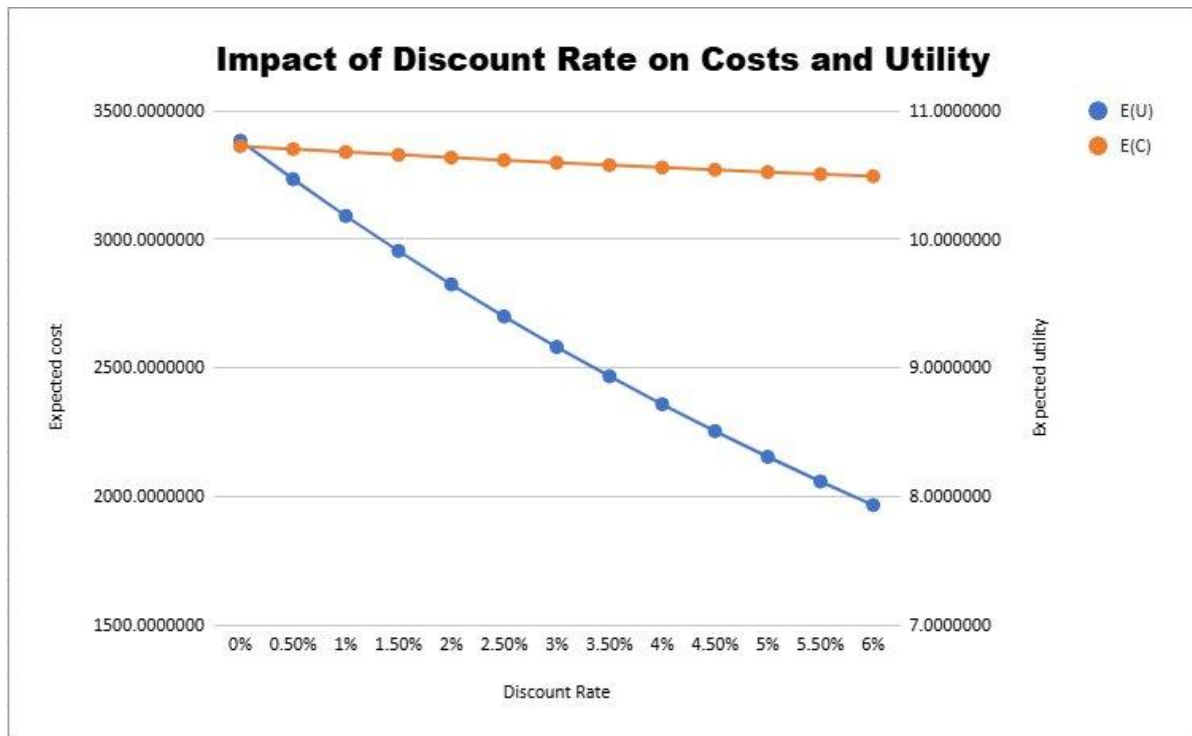


Figure 4: Impact of discount rate on E(C) and E(U) for (ANG,t-,NT);(ANG,t+,CABG).

4.3 Procedural Death Probabilities

Sensitivity analyses of death probabilities is required to assess the safety margin for each ANG and CABG death probability across all strategies, using an incremental probability of 0.0001 for accuracy.

4.3.1 CABG Death Probability

As Table 9 shows, the analysis of CABG death probability favours the base case optimal strategy, with a wide safety margin of around 20x for SVD cases. This means the probability of death from CABG in SVD would need to increase 20 times before the optimal strategy shifts from (ANG,t-,NT);(ANG,t+,CABG) to No Test, MM. However, in MVD, the safety margin is 5x. This reflects the greater severity of MVD relative to SVD (lower QALY) and the fact that the base case $p(\text{death}, \text{MVD}, \text{CABG})$ is higher than in SVD, so the safety margin is smaller relative to SVD. Moreover, when SVD and MVD $p(\text{death})$ are assumed to move together, as expected, the results are more sensitive, with a much lower safety margin: 8.7x for SVD and 3.2x in MVD. In Figures 5, 6, and 7, the line graphs show the shift points from the base case optimal strategy to No Test, MM at 0.60, 0.38, and 0.26 for SVD, MVD, and both, respectively. So, CABG mortality must reach extraordinary values to change the optimal strategy, confirming robustness.

Analysis	Base Case	Switch Point	New Winner	Safety Margin
$p(\text{death}, \text{SVD}, \text{CABG})$ alone	0.030	~0.60	No Test, MM	20x above base
$p(\text{death}, \text{MVD}, \text{CABG})$ alone	0.080	~0.38	No Test, MM	5x above base
$p(\text{death}, \text{SVD} + \text{MVD}, \text{CABG})$ together	both	~0.26	No Test, MM	SVD:8.7x; MVD:3.2x

Table 9: CABG death probability sensitivity.

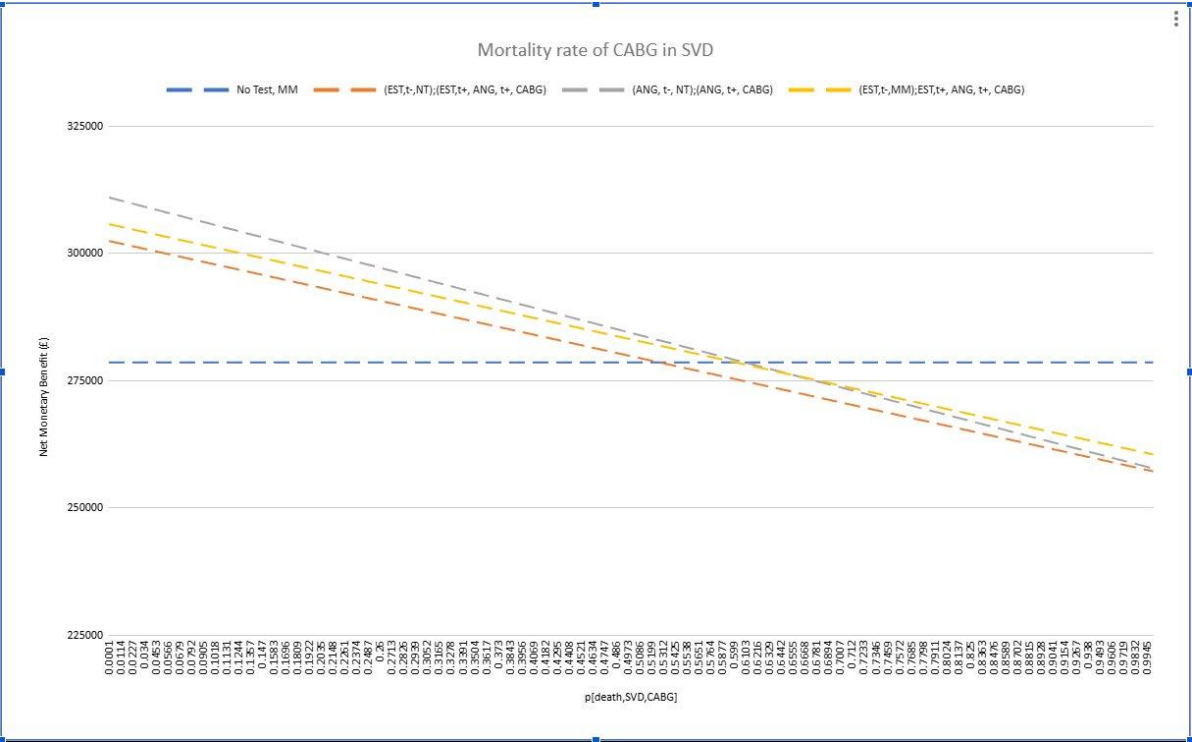


Figure 5: NMB by $p(\text{death}, \text{SVD}, \text{CABG})$.

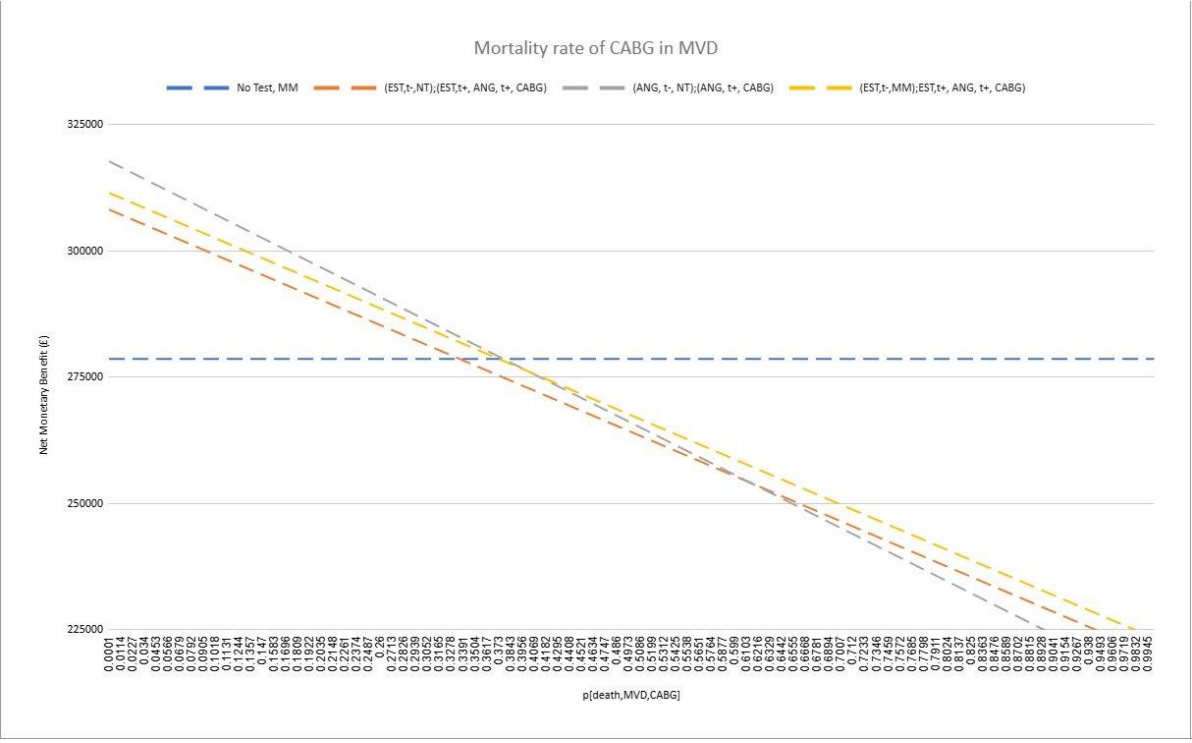
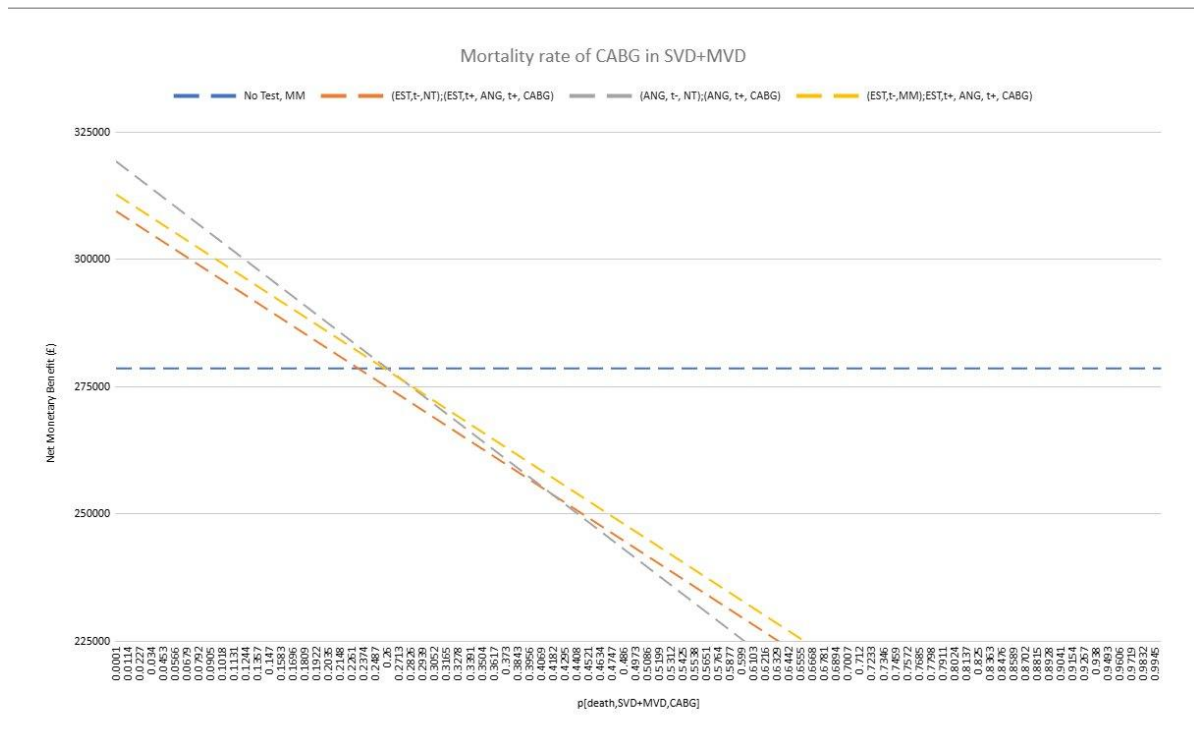


Figure 6: NMB by $p(\text{death}, \text{MVD}, \text{CABG})$.

Figure 7: NMB by $p(\text{death}, \text{SVD}+\text{MVD}, \text{CABG})$ simultaneously.

4.3.2 ANG Death Probability

The ANG analysis shows that the safety margin is even more robust than the CABG one, as *Table 10* shows: 2000x the base case in SVD and 320x in MVD. However, when the probability of death for ANG in SVD, MVD, and healthy patients moves together from 0.0001 to 1, an interesting new optimal strategy appears at as low as 0.03, namely (EST,t-,MM);(EST,t+,ANG,t+,CABG), and then eventually switches again to No Test MM at 0.19.

The reason (EST,t-,MM);(EST,t+,ANG,t+,CABG) is more cost-effective in this case, even though ANG is part of this strategy, which may be to the reason that ANG is performed only on positive EST results, acting as a filter. The MM for negative EST, which covers the remaining 20% of false negatives, may act as a health floor; it improves model robustness under poor surgical conditions, even though it has no effect in the base case. This makes it safer than the base case optimal strategy when the probability of death for ANG is between 0.03 and 0.19. However, after 0.19, the risk becomes too high to use the ANG, and No Test, MM returns as the winner, which is similar to the CABG mortality analysis.

Analysis	Base Case	1st Switch	2nd Switch	Safety Margin
$p(\text{death}, \text{SVD}, \text{ANG})$ alone	0.0003	~0.60 to No Test MM	--	~2,000x
$p(\text{death}, \text{MVD}, \text{ANG})$ alone	0.001	~0.32 to No Test MM	--	~320x
$p(\text{death}, \text{SVD}+\text{MVD}, \text{ANG})$ together	both	~0.21 to No Test MM	--	SVD:700x; MVD:210x
$p(\text{death}, \text{SVD}+\text{MVD}+\text{h}, \text{ANG})$ all	all	~0.03 to (EST,t-,MM);(EST,t+,ANG,t+,CABG)	~0.19 to No Test, MM	50x above ND base
All ANG+CABG deaths together	all	~0.035 to (EST,t-,MM);(EST,t+,ANG,t+,CABG)	~0.13 to No Test, MM	--

Table 10: ANG death probability sensitivity.

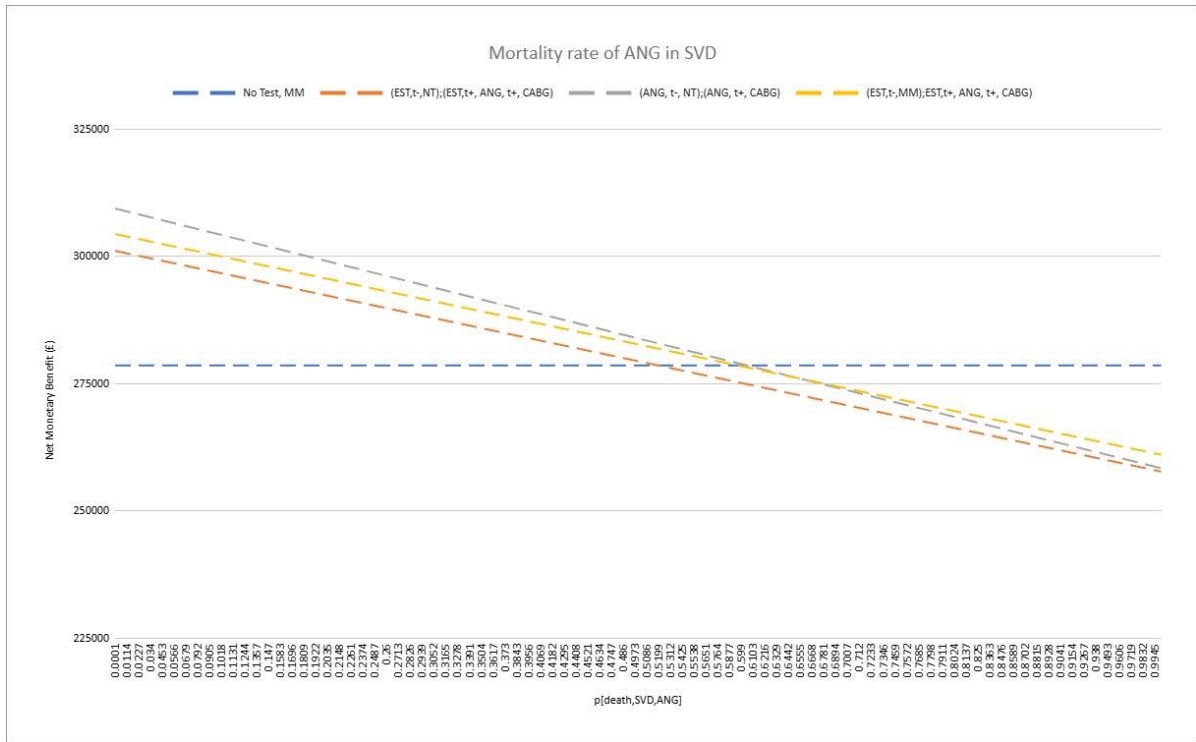


Figure 8: NMB by $p(\text{death}, \text{SVD}, \text{ANG})$.

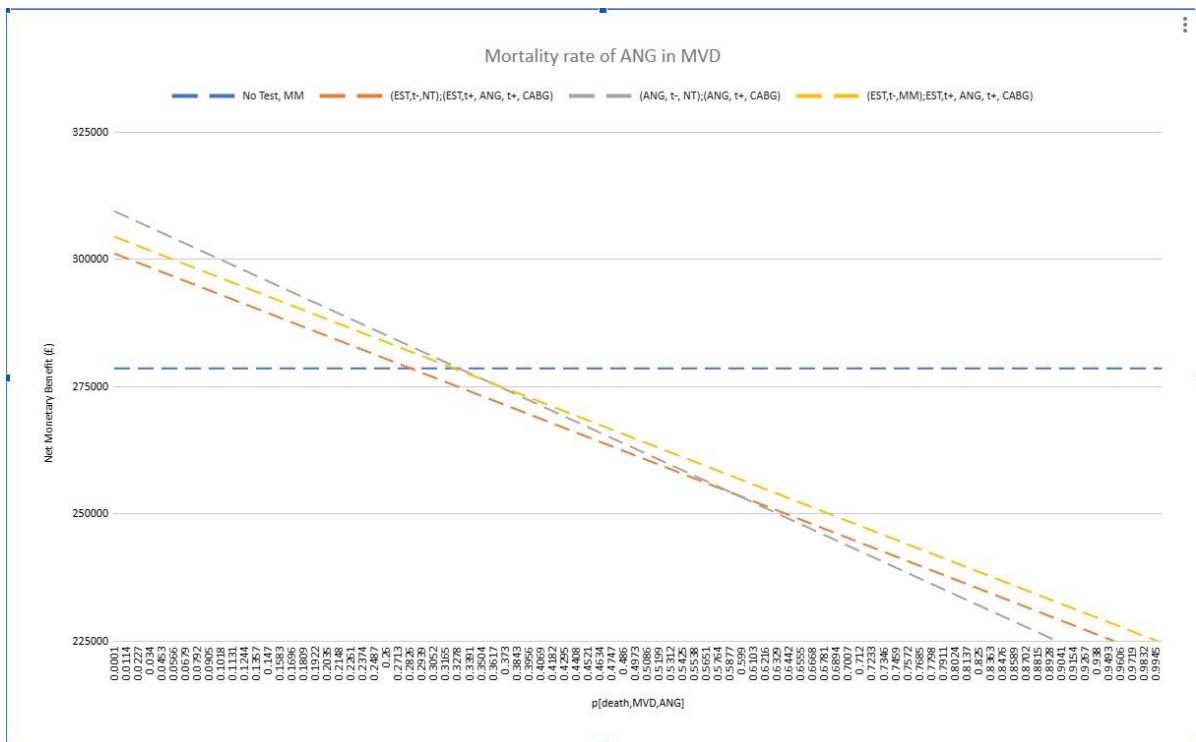
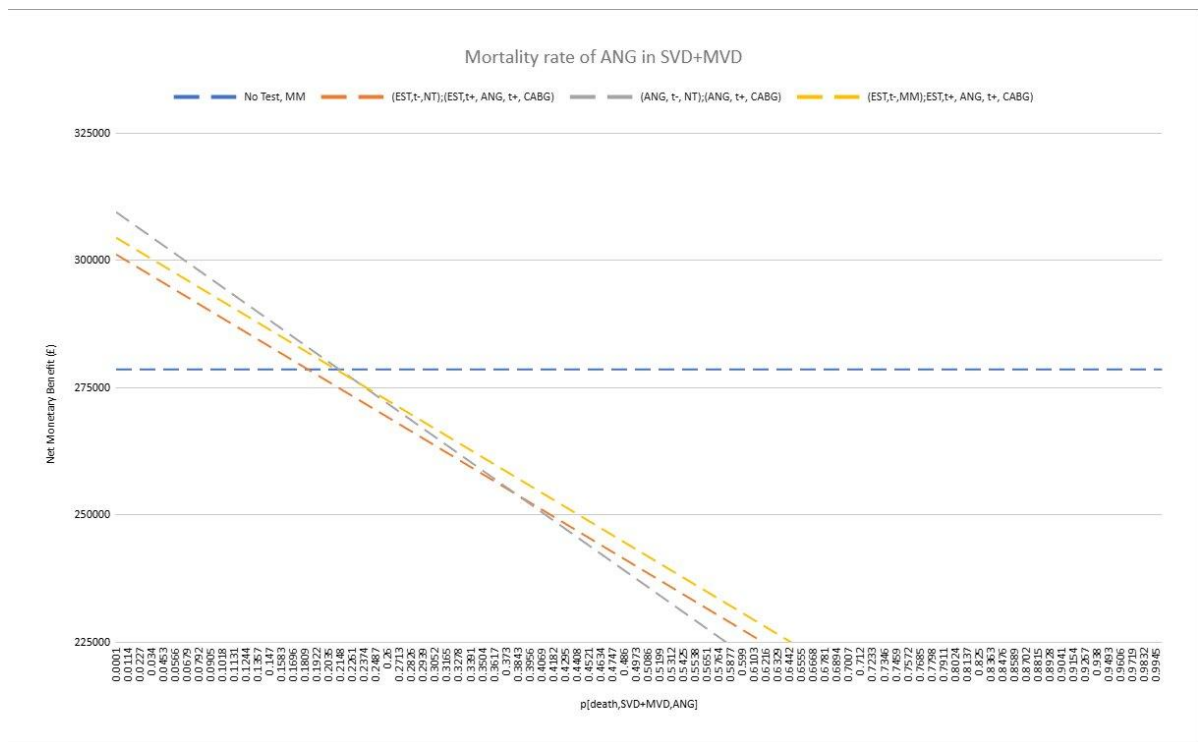
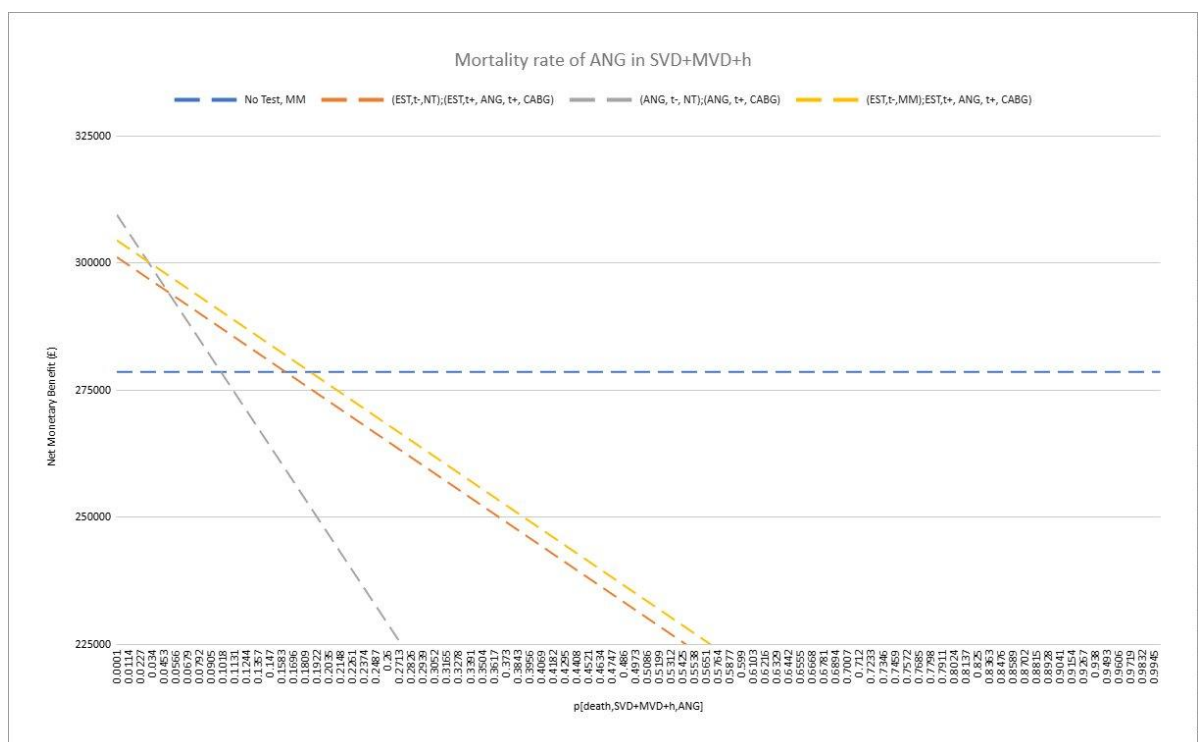


Figure 9: NMB by $p(\text{death}, \text{MVD}, \text{ANG})$.

Figure 10: NMB by $p(\text{death, SVD+MVD, ANG})$.Figure 11: NMB by $p(\text{death, ANG})$ for SVD, MVD and ND simultaneously.

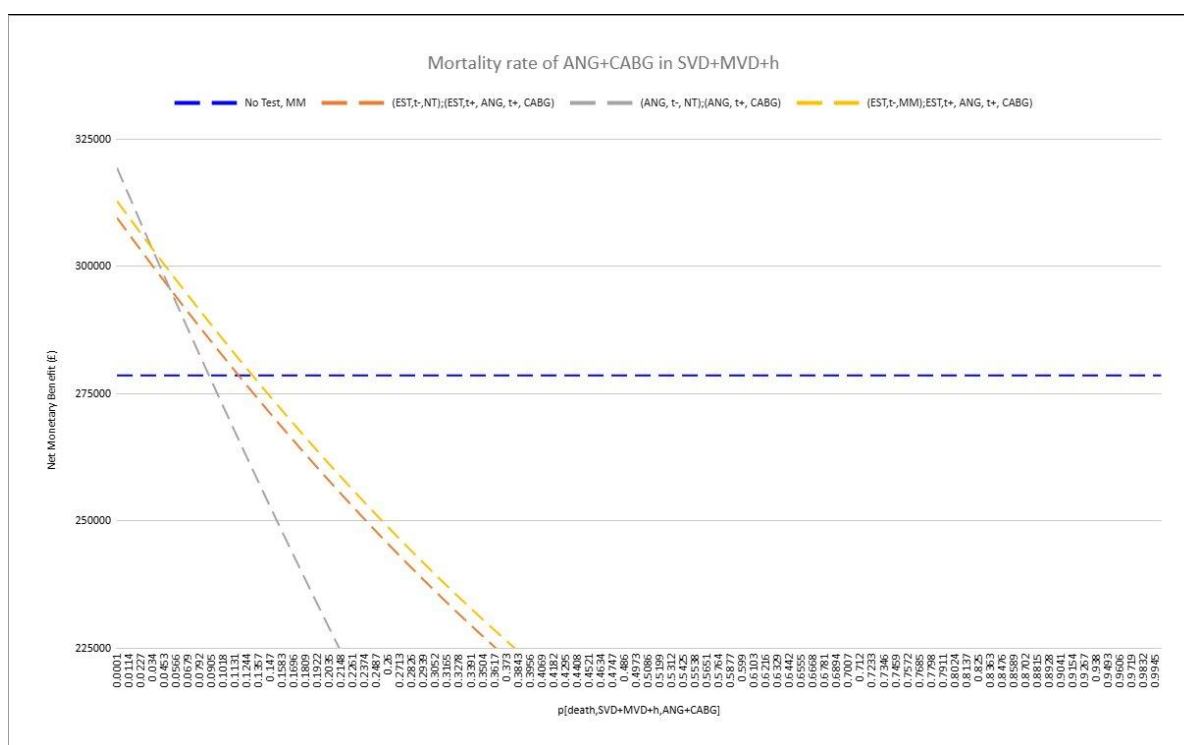


Figure 12: NMB with all ANG and CABG death probabilities varied.

4.4 Value of Perfect Information and Perfect Care

Scenario	Value in Health (QALYs)	Value in Resources (NMB)	Winner
Base case	--	309,415 (reference)	(ANG,t-,NT);(ANG,t+,CABG)
Perfect EST information only	0.000944	247.24 above the reference	(EST,t-,NT);(EST,t+,ANG,t+,CABG)
Perfect care + perfect information	0.806698	28,400.22 above the reference	(EST,t-,NT);(EST,t+,ANG,t+,CABG)
Implied value of perfect care alone	0.80575	~28,153 above the reference	--

Table 11: Value of perfect information and perfect care (WTP=35,000).

Perfect EST information, which assumes EST specificity and sensitivity are 100% and does not change surgical success and mortality rates, yields only 247 NMB and 0.001 QALYs. Moreover, achieving perfect care with perfect information, defined as a zero mortality rate from ANG and CABG, a perfect CABG success rate, and EST specificity and sensitivity, yields 28,400 NMB and 0.807 QALYs. This implies that perfect care alone is worth 28,153 NMB and 0.80575 QALYs.

This means investing more than £247.24 per patient to make the EST test perfect is not cost-effective, and investing more than £28,153 per patient to achieve zero procedural mortality and perfect surgical success rates is also not cost-effective.

These thresholds reflect cost-effectiveness from a health system perspective but do not account for budget impact; such analyses have not been included by us in this analysis because it falls outside the scope of this evaluation. An investment may be cost-effective but exceed the hospital's/country's budget and therefore would be rejected.

5. Discussion

5.1 Summary of Findings

The analysis was conducted according to the Sheffield cardiology department for male patients aged 55 with moderate typical angina, evaluated 15 strategies that combine diagnostic (EST, ANG, No Test) and treatment (MM, NT, CABG) interventions, using ICER and NMB as the main metrics for the base case analysis and NMB as the main metric for the sensitivity analysis. At a WTP threshold of £35,000/QALY and a discount rate consistent with the most recent UK standard from NICE. After removing all strictly and extendedly dominated strategies, three strategies remained in the cost-effectiveness comparison: No Test NT, (EST,t-,NT);(EST,t+,MM), and (ANG,t-,NT);(ANG,t+,CABG). Under UK circumstances, the winning strategy was (ANG,t-,NT);(ANG,t+,CABG), generating an NMB of £309,415 and an ICER of £2,554/QALY relative to (EST,t-,NT);(EST,t+,MM).

5.2 Policy Implications

The policy implications of this analysis can be summarized in three points: first, the ICER of the winning strategy, £2,554/QALY, is well below the threshold, so even after accounting for model uncertainty, it remains low enough that most decision makers would likely accept CABG as cost-effective. However, we should acknowledge that this analysis did not include a budget impact, but it might be safe to assume that the strategy should be accepted by NICE.

Second, ANG/CABG intervention dominated the analysis, regardless of the discounting rate and the threshold (above £2,554/QALY). Moreover, if the mortality rate for ANG and CABG is doubled or tripled, the surgical route would remain the winning way to manage the patients. The only time the EST emerged to win is when it assumed that the EST probability of false positive/negative is equal to zero, and even when this happens, the CABG is still the treatment of choice in this strategy (EST,t-,NT);(EST,t+,ANG,t+,CABG). So, for most cases, even if the discounting rate is as high as 6%, as long as the threshold is above £3024.12, then the winner would still be the ANG/CABG strategy. Lastly, investing in the EST test to make it nearly perfect may not be worth it compared to focusing on the surgical performance to decrease the mortality rate, because a hypothetical zero mortality of ANG and CABG would deliver 114 times greater health returns than investing in a hypothetical perfect diagnostic accuracy (EST)

5.3 Limitations

Key simplifications were flagged inline in Section 2, along with their likely direction of effect. The most significant are: the assumption of a fixed life expectancy, which would be addressed more accurately by a Markov model (Briggs and Sculpher, 1998); uniform treatment irrespective of the ANG-confirmed disease state (SVD or MVD), which diminishes the real value of each strategy; and the uninflated prices of the interventions, which can underestimate the true cost in 2026. The Rosser matrix is included in the case study, but it is prior to current HTA tools such as the EQ-5D-5L.

Only deterministic sensitivity analysis was conducted due to the absence of distributional data, as the case study provided point estimates without standard errors or confidence intervals around parameter estimates, and due to the time constraints on this report, a systematic search of the literature was not conducted to obtain distributional information for each parameter. A probabilistic sensitivity analysis (PSA) would generate a cost-effectiveness acceptability curve (CEAC) by sampling simultaneously from distributions assigned to each parameter, which would provide a comprehensive characterization of parameter uncertainty

The QALYs of the (no-disease/medical management) health state, where the patient got a false positive EST test, or through the No test/MM strategy, is equal to the (no-disease/no treatment) health states, may be considered to be the worst oversimplification in this model because it assumes the MM intervention does not have any side effects or any productivity or financial loss, and ignoring the effect of polypharmacy on the patient who is not even sick.

References

- Goodacre, S. and Calvert, N. (2003) 'Cost effectiveness of diagnostic strategies for patients with acute, undifferentiated chest pain', *Emergency Medicine Journal*, 20(5), pp. 429-433.
- Ruggeri, K., & Nolte, E. (2013). *Pharmaceutical Pricing: The Use of External Reference Pricing*. *Rand health quarterly*, 3(2), 6.
- Briggs, A. and Sculpher, M. (1998) An introduction to Markov modelling for economic evaluation. *Pharmacoeconomics*, 13(4), pp. 397-409.
- Drummond, M.F., Sculpher, M.J., Claxton, K., Stoddart, G. and Torrance, G.W. (2015) *Methods for the Economic Evaluation of Health Care Programmes*. 4th edn. Oxford: Oxford University Press.
- HM Treasury (2022) *The Green Book: Central Government Guidance on Appraisal and Evaluation*. London: HM Treasury.
- Hunink, M. et al. (2001) *Decision Making in Health and Medicine: Integrating Evidence and Values*. Cambridge: Cambridge University Press.
- Gulati, M. et al. (2021) '2021 AHA/ACC/ASE/CHEST/SAEM/NMA/PCNA Guideline for the Evaluation and Diagnosis of Chest Pain', *Journal of the American College of Cardiology*, 78(22), pp. e187–e285. Available at: <https://doi.org/10.1016/j.jacc.2021.07.053>

6 Appendices

Appendix A: Rosser Matrix Valuation

Disability	A None	B Mild	C Moderate	D Severe
I. None	1.000	0.995	0.990	0.967
II. Slight social	0.990	0.986	0.973	0.932
III. Severe social/slight work	0.980	0.972	0.956	0.912
IV. Severe work	0.964	0.956	0.942	0.870
V. Unable to work	0.946	0.935	0.900	0.700
VI. Chair-bound	0.875	0.845	0.680	0.000
VII. Bed-bound	0.677	0.564	0.000	-1.486
VIII. Unconscious	-1.028	*	*	*

Table A1: Williams (1985) Rosser matrix utility values.

Appendix B: NMB Curve Across Discount Rates

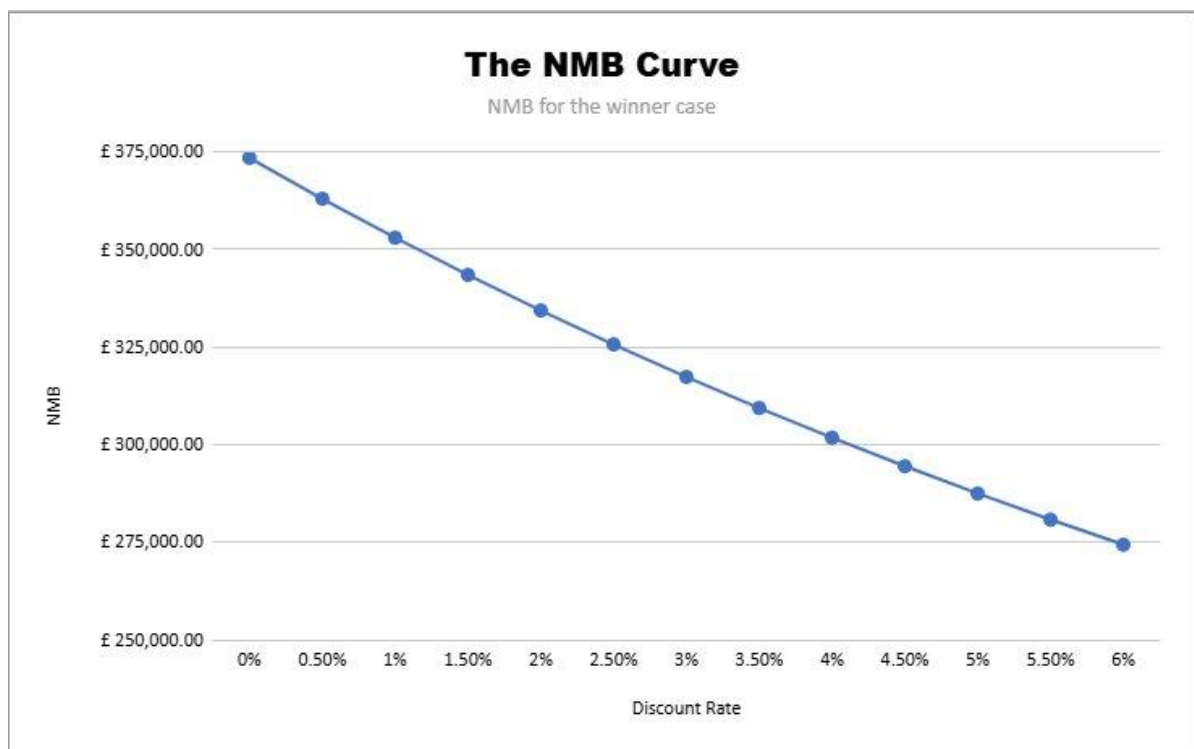


Figure B1: NMB for (ANG,t-,NT);(ANG,t+,CABG) across discount rates 0% to 6%

Appendix C: Full Decision Tree

Figures D1 and D2 present the full decision tree, including all path probabilities, expected utilities, and costs.

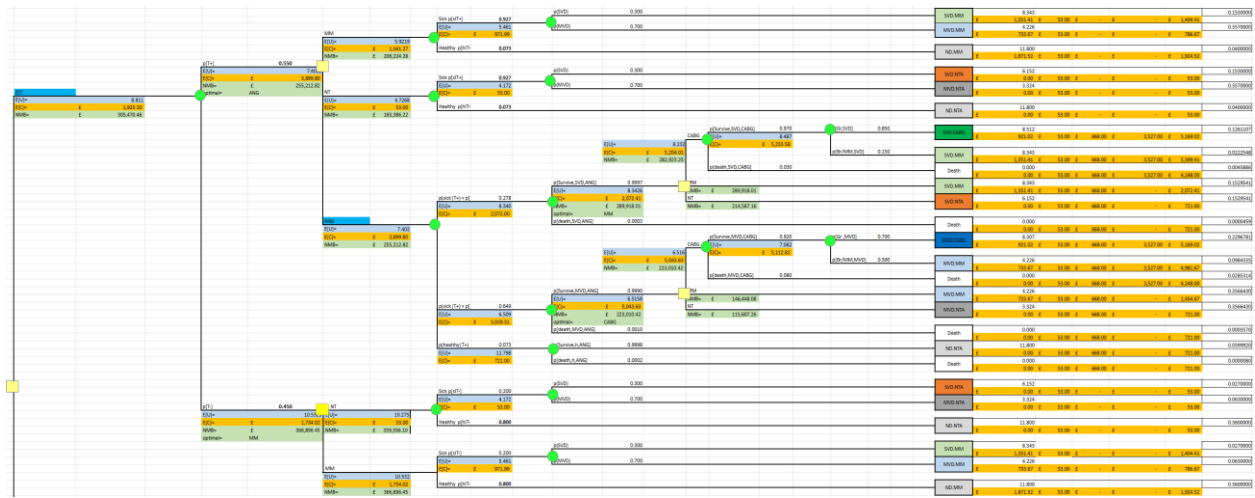


Figure D1: Full EST branch

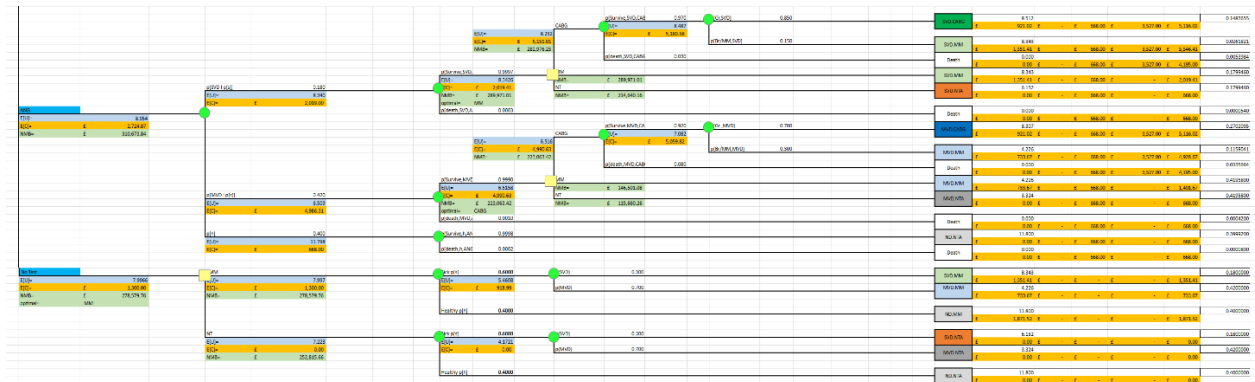


Figure D2: Direct ANG and No Test branches